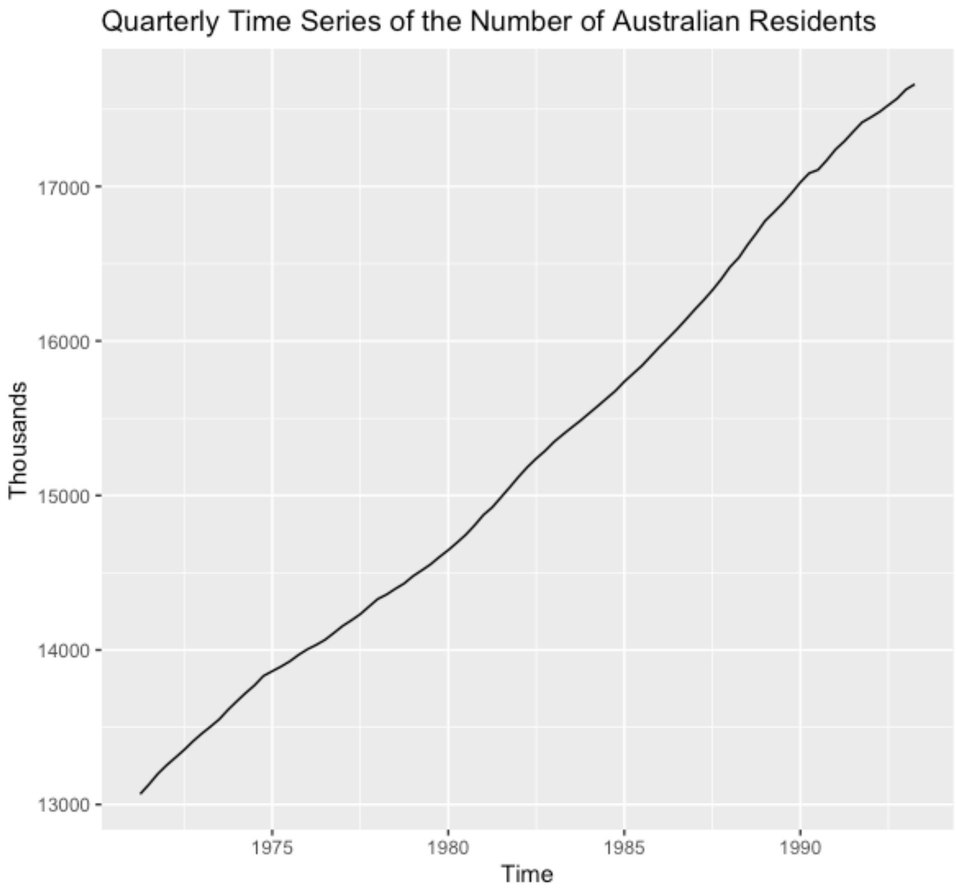
Question 1

Consider the numbers (in thousands) of Australian residents measured quarterly from March 1971 to March 1994 in the time series plot below:



Suppose you decide to fit a Holt's linear trend exponential smoothing model to forecast future values for this time series.

a) Comment on why such a model would be suitable to forecast future values.
   This time series displays increasing linear trend and random variation. Holt's linear trend model is capable of modelling time series with linear trend.
b) What simple forecasting method would serve as a useful benchmark for this time series?
   Drift method.

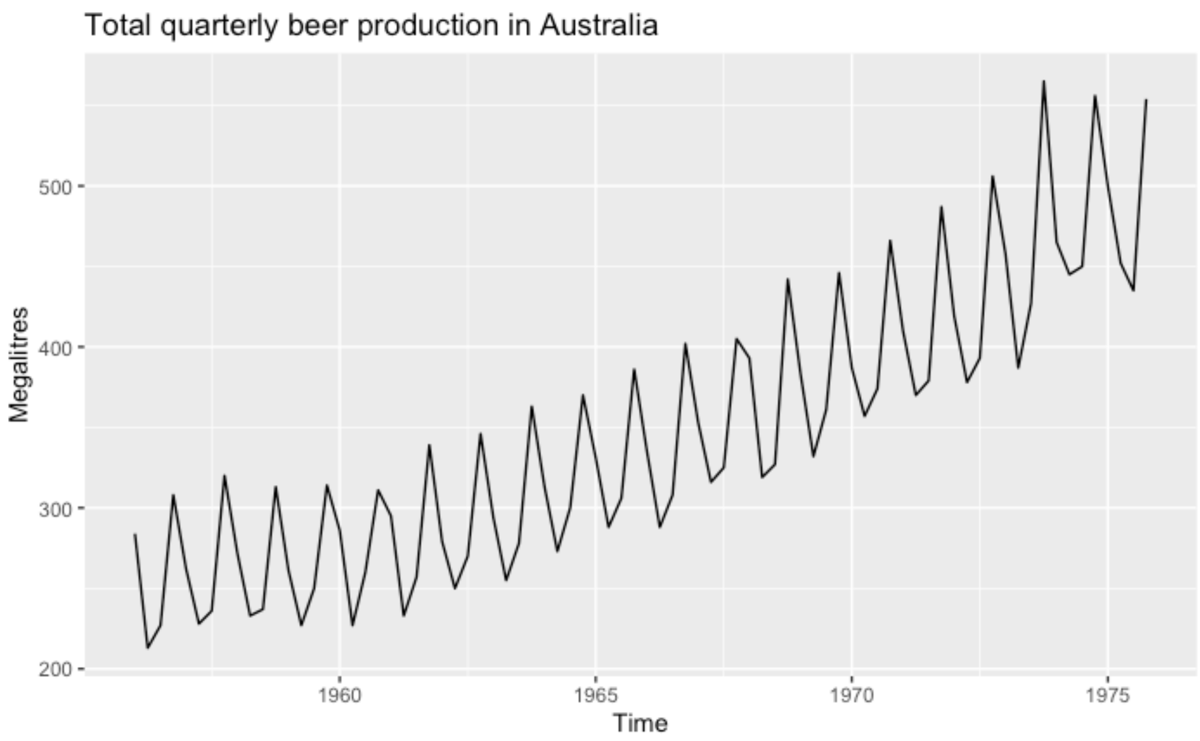
Consider the subset of data, from March 1992 to March 1993 below, with the output from fitting the model as well as fitting a simple forecasting model.

c) Compute the values A-F in the table
   A = 17414.20
   B = 42.79
   C = 17348.9
   D = 17611.45
   E = 17747.34
   F = 17790.26

d) When calculating the level in any given time period, how much weight is placed on the most recent/current level information? Interpret this value.
   0.9999. Almost all weight/importance is placed on the current level information.
e) When calculating the slope in any given time period, how much weight is placed previous slope information? Interpret this value.
   0.5579. About 56% of the weight is placed on previous slope information.
f) Forecast the quarterly values for Jun, Sep and Dec 1993 with the Drift model
   Jun = 17661.5 + 52.2068 = 17713.7068
   Sep = 17661.5 + 2*52.2068 = 17765.9136
   Dec = 17661.5 + 3*52.2068 = 17818.1204
g) If the MAD for the Drift model is 10.26 and for the Holts model is 7.51, what can you conclude?
   The Holts model makes more accurate forecasts, and should be the model we use to make forecasts for this time series.

Question 2

Consider the total quarterly beer production in Australia, from 1956 to 1975



a) What would be the most appropriate simple forecasting method for this time series for short-term forecasts?

   Seasonal naïve model.

b) If our time series only consisted of the data from 1956 to 1960, explain why the simple forecasting method in (a) would produce increasingly inaccurate forecasts in the future?

   The seasonal naïve model can only model seasonality, not trend. Since there is an increasing trend from 1960 onwards, the forecasts from the seasonal naïve model will become more inaccurate over time as the forecasts from the seasonal naïve model will simply be the values in 1959 for each season.

A Holt-Winters Exponential smoothing model was fit to the data from 1968 to 1971. Consider the output from the fitted model, and answer the questions that follow.

c) Compute values A – H (Use the equations on the formulae sheet). Round to 3 decimal places.

   A = 373.916
   B = 3.019
   C = 1.166
   D = 330.974
   E = 424.782
   F = 373.118
   G = 389.696
   H = 495.897

d) Compute the MSE value for this model for the values in the test set.

   42.563